TEAM RUDRAKSHA

SMARTPRICE AI

NEPSE Stock Market Prediction System

A web app that uses machine learning to predict NEPSE stock prices and deliver Buy/Sell guidance.

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Sayapatri Group · AI Project · 2026

01 · PROBLEM

Problem Statement

Emotion-Driven Decisions

Many Nepalese investors trade based on emotion, tips, or rumours rather than data — leading to poor outcomes.

No Localised Tools

Limited access to simple, localised prediction tools designed specifically for NEPSE-listed stocks.

Need for Transparency

Retail investors and students need a lightweight, explainable system to make data-driven decisions.

Our Solution

SMARTPRICE AI is a web-based application that uses 5 years of historical NEPSE data and a trained Linear Regression model to predict the next trading day's prices (Close, High, Low) and provide a clear Buy/Sell signal.

Model Core

Linear Regression built with scikit-learn — simple, fast, and interpretable for students.

- 5 Years of NEPSE Data
- 1,149 Rows / Stock

Web App

Streamlit UI for easy interaction. Enter today's inputs, get instant predictions & guidance.

- Buy / Sell Signal
- Dark Theme UI

How the System Works

01 **Collect Data**

5 years of NEPSE historical data for 3 stocks

02 **Compute Features**

MA-5, MA-10, Momentum, Daily Range, Prev High/Low

03 **Train Model**

Linear Regression — 80/20 train/test split

04 **Predict & Signal**

Output Close, High, Low + Buy/Sell recommendation

Dataset & Feature Engineering

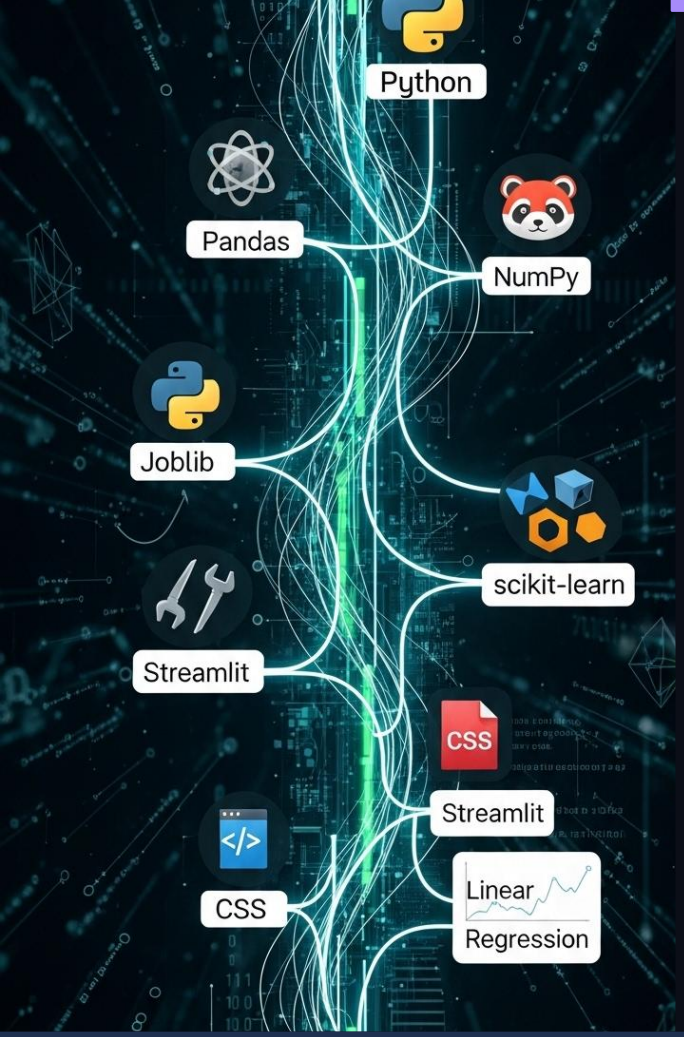
NABIL	NTC	CIT
Nabil Bank	Nepal Doorsanchar	Citizen Inv. Trust
1,148 rows	1,149 rows	1,149 rows
0.91 R^2	0.87 R^2	0.94 R^2

Engineered Features (7 total):

- PREV_CLOSE
- MA_5
- MA_10
- MOMENTUM
- PREV_HIGH
- PREV_LOW

5 YEARS

Datas for training



05 · TECHNOLOGIES

Technologies Used

Python	Core programming language
Pandas & NumPy	Data cleaning and numerical feature computation
Scikit-learn	Linear Regression model for price prediction
Joblib	Save and reload trained models efficiently
Streamlit	Quick, interactive web interface for live demos
HTML/CSS	Custom UI styling for a modern AI look

Features of SMARTPRICE AI

Price Predictions

Predicted Close, High and Low for the next trading day.

Buy/Sell Signal

Simple action recommendation based on predicted movement.

Feature Set

MA (5,10), Momentum, Daily Range, Previous Highs/Lows.

Historical Coverage

Uses 5 years of NEPSE data to capture seasonality and longer-term trends.

Model Performance Results

Algorithm

Linear Regression

Train/Test: 80% / 20% · shuffle = False

0.94
Best R^2

99%+
Accuracy

3.12
Min MAE

3
Stocks

Stock	MAE	RMSE	R^2 Score	Accuracy
Nabil Bank	3.12	5.06	**0.9104**	99.38%
Nepal Doorsanchar	5.48	8.15	**0.8724**	99.37%
Citizen Inv. Trust	11.50	18.98	**0.9395**	99.39%

Predicted Low: 748.50

Live Demo Explanation

1. Select stock (e.g., Nabil Bank) and input today's close and optional technical indicators.
2. Click Predict — Streamlit sends data to the loaded Joblib model instantly.
3. Model returns Predicted Close, High, Low and a clear Buy/Sell signal.
4. Results shown with confidence range and a brief explanation for learners.

Designed to be fast, interpretable and easy to use during the live presentation.

Future Improvements

Deep Learning

Upgrade to LSTM/Transformer models for capturing temporal patterns.

Real-Time NEPSE API

Connect to live NEPSE feed for streaming predictions and alerts.

Expand Coverage

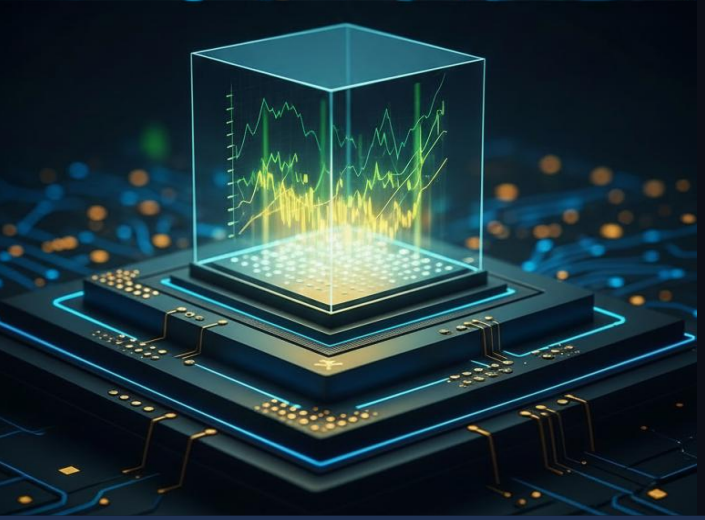
Add more NEPSE-listed companies and instrument-level models.

Visual Dashboards

Live charts, confidence intervals, and backtest result visualisations.

Ensemble Methods

Improve accuracy by combining multiple ML algorithms.



Conclusion

SMARTPRICE AI demonstrates a practical, student-built application of machine learning to Nepal's stock market. With 5 years of NEPSE data and an accessible web UI, it helps learners and retail investors make more informed decisions.

Website link: https://smartpriceproai.streamlit.app

3	5yr	0.94	99%+
Stocks	Data	Best R^2	Accuracy

Thank You!

Questions are welcome.

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